

MSc INDIVIDUAL PROJECT

IMPERIAL COLLEGE LONDON

DEPARTMENT OF BIOENGINEERING

**[CONFIRM: title] Teleoperation of a
Backpack-Mounted Dual-Arm Supernumerary
Robotic Limb System**

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CONFIRM: project title, author name, supervisor and co-supervisor on the title page, and the Imperial logo (run `epstopdf title/logo.eps` once ghostscript is installed).

Abstract

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Acknowledgements

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Chapter 1

Introduction

Supernumerary robotic limbs (SRLs) are additional arms worn on the body, which act alongside the wearer's own limbs rather than replacing or rehabilitating them. They differ from prostheses in that the wearer retains full natural function, and from industrial manipulators in that the base of the robot is a person: it accelerates, sways and steps, and it is soft, valuable and irreplaceable. The work reported in this thesis concerns a wearable system of two seven-degree-of-freedom Kinova Gen3 arms carried on a backpack frame and teleoperated from an instrumented mannequin master arm.

The configuration that this system implements is deliberately a *two-person* one: the arms are worn by one individual and commanded by another. That separation is the source of most of the interesting engineering difficulty in the project, because the person who bears the risk has no control input, and the person issuing the commands has neither proprioception of the wearer nor any vestibular sense of the base motion they are working against.

1.1 Objectives

The project set out to build and characterise a complete teleoperation stack for this configuration, and specifically to establish, by measurement rather than by demonstration, what such a system can and cannot do. The objectives were:

1. to implement an end-to-end pipeline from the master arm's sensors to joint trajectories on two arms, including the mapping from the operator's limb pose to a commanded end-effector pose;
2. to design a safety layer appropriate to a robot mounted beside a person's head, and to demonstrate its behaviour under injected faults rather than only under nominal conditions;
3. to characterise the reachable workspace of the mounted configuration, and to determine which manipulation tasks it can actually support;
4. to establish what performance survives when the master arm's sensing degrades, since a substantial fraction of its channels are unreliable;
5. to design an experimental protocol capable of separating the contribution of machine autonomy from that of the human operator.

1.2 Challenges

Three difficulties shaped the work more than any others.

The first is that *the base moves*. Conventional manipulator control assumes an inertial base; here the base is a standing person. This is a recognised problem in the supernumerary-limb literature but it is normally treated in the single-person case, where the wearer is also the operator and their own motor commands are available as feed-forward. In the two-person case that information does not exist.

The second is that *the input is degraded*. The master arm's potentiometer channels are, in their present physical state, substantially unreliable, and the system therefore has to remain usable on a subset of its intended sensing rather than assuming all of it.

The third is that *almost every failure in this system is silent*. An arm that has stopped moving looks the same whether it is blocked for a good reason, blocked for a bad one, or not receiving data at all. A great deal of the engineering effort described in this thesis went into making the system's internal state legible, and a recurring theme of the results is that measurement instruments failed more often, and more quietly, than the system being measured.

1.3 Contributions

The contributions claimed here are limited to those the implementation and its recorded artefacts can support.

1. **A complete, safety-instrumented dual-arm teleoperation stack** for a backpack-mounted pair of Kinova Gen3 arms, comprising master-arm sensing, a spherical pose mapping, collision-aware inverse kinematics with redundancy resolution, a clearance floor referenced to the wearer's body, and a multi-path emergency stop. The behaviour of the safety layer was exercised by systematic fault injection rather than by inspection.
2. **A workspace characterisation of the mounted configuration**, including the negative result that the two arms share no reachable workspace, which blocks inter-arm handover for geometric reasons that no control or planning change can address.
3. **A quantified account of degraded-input teleoperation**: which capabilities survive the loss of individual sensing channels, and what each loss costs.
4. **A verification methodology**, with a set of recorded runs covering every task, scenario and autonomy condition, checked automatically for object handling, gripper actuation and wearer clearance.
5. **An experimental design for the two-person configuration**, including measures for the wearer as a participant rather than as apparatus.

Chapter 4 states explicitly, for each result, whether it was verified in simulation, exercised only against mock hardware, or validated on physical arms.

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Chapter 2

Background

2.1 Supernumerary robotic limbs

Supernumerary robotic limbs augment rather than restore: the wearer keeps their own limbs and gains additional ones. Reported applications span bracing and postural support, supernumerary fingers, and general-purpose extra arms for shared-workspace tasks.

Two questions recur in this literature and both matter here. The first is how the extra limbs are *commanded*, given that the wearer's own limbs are already occupied; reported approaches include body-mounted sensing of residual degrees of freedom, foot and tongue interfaces, and inference from the wearer's ongoing motion. The second is whether the wearer comes to experience the limbs as part of their body, which is normally framed in terms of body schema and ownership.

The configuration used in this project sidesteps the first question entirely by placing the operator outside the wearer's body, and makes the second inapplicable: the arms are never mechanically linked to the wearer's own limbs, so ownership language does not apply to them.

2.2 Teleoperation with a separate wearer

The direct predecessor to this configuration is Fusion, which mounted a robotic head and two arms on a backpack worn by one person and drove them from a second, remote operator wearing a head-mounted display. Fusion defined three levels of what it termed bodily driven communication, distinguished by how much the robot arms were physically coupled to the wearer's own arms; in all three the operator commanded the motion and the machine decided nothing. It was presented as an Emerging Technologies demonstration and reported no user study or quantitative evaluation, so it establishes the architecture as buildable rather than characterised.

More recently, the spatial behaviour of back-mounted arms around a wearer has been studied directly, with a concealed separate operator and varying levels of apparent autonomy. That work reported that higher autonomy did not improve the wearer's perceived safety, and produced body-zone guidance on approach distances and trajectory shapes. It is relevant to the present system both as a precedent for the two-person arrangement and as a caution against assuming autonomy is experienced as reassuring.

2.3 Teleoperation from a moving base

Where the manipulator base is not inertial, base motion enters the control problem as a disturbance. Approaches reported for mobile manipulators include disturbance estimation and compensation, and master–slave schemes with local feedback on the follower. For supernumerary limbs specifically, the wearer’s body is the floating base and the disturbance is the wearer’s own postural sway. In the reported cases the wearer is also the operator, so the disturbance is self-generated; the two-person configuration removes that, which is the distinction this project is positioned against.

2.4 Shared autonomy in teleoperation

Assistive teleoperation is commonly formulated as an arbitration between the operator’s input and a robot policy derived from a prediction of their goal. A structural weakness of predicting before acting is that confidence often arrives too late to be useful, which motivates formulations that assist under goal uncertainty.

2.5 Platform: Kinova Gen3, ROS 2 and MoveIt 2

The arms used are seven-degree-of-freedom Kinova Gen3 manipulators with Robotiq 2F-85 grippers. The redundant seventh degree of freedom is used in this work for collision avoidance: the elbow can be swung about the shoulder–wrist axis without moving the gripper, which provides a family of solutions to search when a pose is otherwise blocked.

The software stack is ROS 2 Jazzy with MoveIt 2 for kinematics and collision checking. Inverse kinematics is solved by TRAC-IK. TRAC-IK’s use of randomised restarts is relevant to the verification methodology described in Chapter 3, because it makes a single successful solve a weak statement about reachability.

2.6 Open-vocabulary visual grounding

The autonomy modes in this system require objects to be located from a description rather than from a fixed model. Open-vocabulary detectors admit an arbitrary class list at inference time. Speech transcription uses a Whisper-family model. A deliberate decision not to use an end-to-end vision–language–action model is discussed in Chapter 5.

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Chapter 3

Method

3.1 System architecture

The system is organised as six ROS 2 packages with a deliberately one-way dependency structure. `srl_teleop` contains teleoperation only and depends on nothing else in the workspace; `srl_perception` contains object detection and pose estimation; `srl_autonomy` depends on both and adds grasp generation, intent inference and handover arbitration; `srl_experiments` depends on all of them and contains the protocols, logging and analysis. `srl_description` holds the robot and wearer model and `srl_moveit_config` the planning configuration.

The direction of that dependency arrow is load-bearing rather than tidy. Direct teleoperation is the baseline condition against which every autonomy condition is compared, so if the teleoperation package imported anything from the autonomy package the comparison would be circular. The property was checked in two ways: by searching for imports, and by physically removing the autonomy and perception packages from the install tree and confirming that the teleoperation follower still reached its running state.

3.2 Master arm and sensing

The operator commands the system through an instrumented mannequin arm. Joint angles are read by potentiometers and limb orientation by an inertial measurement unit, both sampled by a Teensy 4.1 microcontroller and streamed over USB serial as fixed-format frames.

The serial port is discovered rather than configured. The board enumerates as either `/dev/ttyACM0` or `/dev/ttyACM1` depending on the order of USB attachment, so the driver globs the candidate devices, opens each and sniffs for a valid frame header. On failure it raises an exception naming the recovery command rather than continuing silently, because a node that runs with no data is indistinguishable from one with a dead sensor.

Access to the port is made exclusive with an advisory lock taken on the open file descriptor. This is not merely defensive: the port *search* itself reads bytes, so probing a port that another node already owns removes frames from that node's stream. Two readers on one board previously invalidated a full day of recordings.

3.2.1 Pose mapping

The commanded end-effector position is constructed in spherical form about the master's shoulder rather than by forward kinematics of the full chain. Elevation is taken from the wrist inertial measurement unit, as the angle between the limb's long axis in the sensor frame and the measured gravity vector; this is drift-free and requires no integration. Azimuth is taken from the shoulder roll potentiometer alone, zero-referenced at calibration. Reach

magnitude is taken from the forward kinematics of the proximal joints, using magnitude only.

The reason for this decomposition is that the full seven-joint forward kinematic solution collapsed directionally when it was evaluated against recorded gestures: a singular value decomposition of the resulting positions placed most of the variance on a single axis. The spherical decomposition isolates the three quantities the pipeline actually needs and allows each to be sourced from the most reliable available sensor.

The azimuth channel carries a known limitation. The shoulder roll axis is fixed in the mount, so rolling it moves the tip only once a downstream joint has offset the tip from the roll axis; azimuth per degree of roll therefore depends on how bent the arm is, and azimuth and elevation are not independent as the spherical model assumes.

3.2.2 Degraded operation

Because a substantial fraction of the potentiometer channels are unreliable, the system treats channel loss as an expected condition rather than a fault. Validation is scoped to the channels the active mapping mode consumes, so a dead joint that the mode does not use cannot reject an otherwise-good frame. A channel manager publishes which channels are live and what each loss costs, and can retire a channel that has been reporting a firmware-clamped constant.

The policy is to degrade rather than to fail, with one exception that is stated explicitly in the implementation: if no positional channel survives, position is reported as unavailable rather than substituted. Inventing a position for a robot mounted beside a person's head is the most dangerous available failure mode.

3.3 Kinematics and conventions

3.3.1 Coordinate convention

The world frame used throughout has x lateral, y forward with the wearer facing $+y$, and z up. This is *not* the usual ROS convention of x forward, and the difference caused a substantial error in the mount orientation which is described in Chapter 4. The frame was confirmed independently from the geometry of the wearer model itself.

A separate convention conversion is required between the Kinova controller, which reports actuator positions in degrees over $[0, 360)$, and the ROS joint state in radians over $[-\pi, \pi)$. This is implemented in a single module, `kortex_convention`, and is unit-tested, because it is applied on every homing command and an error in it is a large unattended motion.

3.3.2 Inverse kinematics and the redundancy

Commanded poses are solved with TRAC-IK through MoveIt's `compute_ik` service, with collision checking enabled against a model that includes the wearer. The seeding is taken from the current joint state, so the solution is continuous with the arm's present configuration.

Two mechanisms handle failure. On an unsolved pose the redundant degree of freedom is re-seeded and the solve retried, fanning the elbow either side of its natural posture; this recovers poses that are reachable but blocked in the posture the arm happens to be in. On a solution far from the current state the follower commits a bounded step towards it rather than discarding it. The distinction matters: an earlier implementation discarded any solution further than a threshold from the current state, which could never converge, because a discarded solution leaves the arm where it was and the next solution is equally far away. Solutions are rejected only when the target pose

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has barely moved and the solution has jumped, which indicates a redundancy flip, and even then a counter forces acceptance after a bounded number of consecutive rejections so that rejection cannot deadlock either.

3.4 Wearer model and clearance

The wearer is represented in the robot description as a kinematic model with collision geometry for the torso, head, hips, legs, arms and hands. Collision checking against this model is what makes `avoid_collisions` meaningful for the operator rather than nominal.

Pairs of links that are permanently adjacent because they are bolted together are excluded in the semantic robot description; everything distal of the upper arm remains enabled, since that is what must actually avoid the wearer. An exclusion silences an alarm without moving any metal, and the mount interference described in Chapter 4 is recorded as a mechanical action rather than treated as resolved by the exclusion that hides it.

Separately from the binary collision check, a geometric clearance is computed from the transform tree between the arm's distal links and primitive approximations of the wearer's body. This produces a margin rather than a contact flag, and the follower refuses to publish below a configurable floor. The distinction between a contact test and a margin is central to one of the results in Chapter 4.

3.5 Safety layer

3.5.1 Emergency stop

The emergency stop has three independent triggers, any of which latches until an explicit reset: a service call or topic message; both master buttons held simultaneously for a sustained interval; and a dead-man on the master data stream. On trip, each arm's measured joint positions are republished as a short trajectory, so a follower that continues commanding is continuously overridden.

The dead-man keys on the *source timestamp* of the master pose rather than on message arrival. This is the single most important detail in the safety layer, and Chapter 4 describes why: the master node does not go silent when it degrades, it republishes its last good value, so an arrival-based dead-man cannot fire on the failure that actually occurs.

3.5.2 Making blocking visible

Several mechanisms in the system can legitimately stop motion. The recurring defect was never that they blocked, but that blocking was indistinguishable from working. Every such mechanism therefore registers with a monitor that requires a recovery string at registration, announces the first block by name, escalates if a block is sustained, and publishes an aggregated state.

The monitor treats a blocker that has not been re-asserted as *expired* rather than clear. A held condition is re-asserted every cycle by design, so a blocker nobody is asserting indicates that the asserting loop has stopped, and the guarded condition is therefore unknown rather than absent. Consumers treat expired as blocking.

3.6 Simulation and control interface

All results reported in this thesis were obtained against the simulated stack, in which the hardware interface is `mock_components/GenericSystem` and joint commands are echoed with no dynamics. Trajectories are sent to `joint_trajectory_controller` instances, one per arm and one per gripper.

A separate high-level bridge exists for the physical arms, using the Kinova high-level velocity API rather than the cyclic real-time interface.

3.7 Verification methodology

Because much of this work consists of measurements rather than features, the method by which measurements are trusted is itself part of the method.

Two rules were adopted after repeated failures. The first is that a pose which passes a single inverse-kinematics call is not established as reachable: TRAC-IK restarts randomly, so a pose at the edge of the feasible set is a coin flip and one call is one flip. Verification is therefore N repeats over a path densified to a fixed spatial step, rather than a single call at each declared waypoint, since the arm travels the segments between waypoints and those are not reachable by virtue of their endpoints being reachable.

The second is that an instrument must be validated against a known answer before a surprising measurement taken with it is reported. Every analysis routine that will touch experimental data has a test that feeds it an input whose answer is known by construction and confirms that it is recovered.

3.8 Recorded verification

Each task, scenario and autonomy condition is driven in simulation by a scripted operator and recorded in two forms. The first is a rendering generated directly from the transform tree, drawn from the same samples that produce the logged metrics, so the picture and the numbers cannot disagree. The second is a screen capture of the RViz visualisation from four fixed camera angles, which is what a human reviewer actually watches.

Task objects are published as visualisation markers at their physical dimensions and are attached to the gripper only while the measured gripper joint indicates closure on an object, so that the rendered object follows the gripper rather than standing in for it.

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Chapter 4

Results

4.1 Scope of the evidence, and how to read this chapter

Every result in this chapter is labelled with the strength of evidence behind it, using three tiers:

Simulated measured against a running planning and control stack with a mock hardware interface;

Mock-exercised the code path ran and its failure modes were injected, but no physical device was involved;

Hardware a physical arm, a physical microcontroller or a human participant took part.

No result in this chapter is of the Hardware tier. No experiment was run with a human participant, and no logged artefact from a physical arm exists in the project repository. The engineering log records real-arm bring-up activity, but as no corresponding data file was retained, those figures are not reported here as measurements of this work.

4.2 Workspace and reachability

4.2.1 Continuous reachability from the home pose

Reachability was measured as a swept extent rather than as pointwise feasibility, because teleoperation requires the arm to travel to a pose from where it already is, without a branch change. Twenty-six directions were swept in one-centimetre steps from the home configuration, each step seeded from the previous solution.

Repeating the sweep ten times gave identical quartiles: the inter-quartile range was 0.000 m on every direction for both arms. The measurement is therefore repeatable. What was *not* repeatable in earlier runs was the summary statistic: isotropy computed as the minimum over directions of a single sweep is dominated by rare early terminations, and varied by a factor of two between runs. On medians it is stable, at 0.16 for the left arm and 0.10 for the right. (*Simulated; recordings/baselines/workspace_n10_20260806.json*)

The early terminations were traced to the measuring tool rather than the system. A sweep is a chain of up to ninety solver calls and any single failure ends the direction; the follower does not suffer this because it re-seeds the redundant joint and retries. Giving the sweep the same retry reduced early terminations from nine to zero on the left arm and three to zero on the right, while the medians barely moved — which is what confirms the outliers were chain compounding rather than geometry.

4.2.2 The two arms share no workspace

Over a grid of sixty-three cells spanning the frontal working volume, no cell was reachable by both arms. With collision checking disabled only three cells were shared, and the wearer's own geometry removed all three. (*Simulated; scripts/verify_task_scenes.py.*)

This is a geometric property of the mount and the parked joint configuration, not a planning or control limitation, and it is not addressable by a wearer reposition, because moving the base moves both arms together and leaves their mutual separation unchanged. The consequence is that any task requiring the two arms to meet at a point — an inter-arm handover — cannot be performed in this configuration.

A parametric sweep over candidate mount placements indicated that a buildable change, reducing the separation of the two mounts, would raise the number of shared cells from three to twenty-two. (*Simulated; recordings/baselines*) The sweep is purely kinematic: it moves the arms' bases without moving the wearer, so its figures are an upper bound and any candidate requires a full clearance check before it is adopted. It was recorded and not applied, because changing the mount invalidates the home pose, the workspace anchor and every clearance figure in this chapter.

4.3 Master-arm channel health

Fourteen potentiometer channels were classified from a recorded capture. The classification distinguishes four states, of which one had to be introduced during the analysis: a failing channel does not fall silent, but returns values spanning a plausible range that are not a trajectory, with consecutive updates unrelated. Range alone calls that healthy.

Six of the fourteen channels were usable. (*Simulated analysis of recorded data; recordings/baselines/channels_202*

Two prior conclusions were overturned by this analysis, both because the recorder sampled at approximately 50 Hz while the sensors update at 14.7–17 Hz, so 82–88% of adjacent rows are bit-identical. Any statistic computed over consecutive rows is therefore dominated by duplicates: the apparent noise floor of 0.000 on every channel is an artefact of the recorder, and the zero-spread signature previously used to identify a dead channel is the same artefact. Computed between distinct sensor updates, the noise floor is 0.2–5.4°.

4.3.1 Cost of degraded operation

With the incoherent channels frozen at their reference values, the contribution of those channels alone to the commanded tip position was measured by replaying recorded traces with every coherent channel held at its median. The left arm's incoherent channels contributed 54.5 mm of commanded tip motion from a perfectly still hand; the right arm's contributed 0.014 mm. (*Simulated replay of recorded data; scripts/prove_degraded_mode.py.*)

Regressing the true reach on the surviving channels showed that the left arm retains no reach observable at all once its shoulder and elbow channels are excluded ($R^2 = 0.133$, residual 93% of the true spread), whereas the right arm does ($R^2 = 0.986$). This identifies the two specific channels whose repair would restore the most capability.

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4.4 Safety layer behaviour

4.4.1 Fault injection

Fourteen distinct faults were injected and all fourteen were handled, meaning the fault was detected, the system responded in the designed way, and the response was visible in the published state. (*Mock-exercised; recordings/fault_acceptance/.*)

Faults were also injected mid-trial into all six experiment runners. In each case the affected trial was marked invalid with the cause recorded, the session continued, and the analysis excluded the trial as pre-registered. A session in which every trial is invalidated fails loudly rather than analysing to an empty result, which previously was indistinguishable from a null finding.

4.4.2 The dead-man defect

The most consequential defect found in the safety layer was that the dead-man could not detect the failure that actually occurs. The master node does not fall silent when its input degrades: it substitutes its last good value and continues publishing at full rate. Because it stamped that substituted message with the current time, neither the topic nor the timestamp went stale, and a dead-man keyed on message arrival could not fire.

Only a clean cable removal produced silence, which is why every cable-pull test had passed. The test and the failure mode were different failures. After keying the dead-man on the source timestamp of the data, the trip times measured were 0.94–0.99 s for a degraded-but-publishing master and 0.97 s for a silent one, against a deadline of 1.00 s, with no false positive over 243 fresh frames. (*Mock-exercised.*)

4.5 Recorded verification of the task set

Every task, scenario and autonomy condition was driven in simulation and recorded, giving eighty-seven runs. Each was checked automatically for content, for object handling and for gripper actuation.

Across all eighty-seven runs, no arm link entered the wearer's collision geometry, and no run showed a stationary robot. (*Simulated; recordings/verification/.*)

4.5.1 Objects are carried, not merely rendered

The check that the manipulated object travels with the gripper was made from the recorded video rather than from the code that drew it, on the grounds that the marker pose is computed from the live end-effector pose and would therefore track by construction. For the pick-and-place task the evidence requires no centroid tracking at all: the count of pixels belonging to the supply of blocks falls as blocks are removed, while the count belonging to placed blocks inside the container rises. A gripper closing on nothing changes neither.

4.5.2 A defect found by watching, not by testing

The gripper was never commanded in the first round of recordings. The mock hardware interface initialises the driven knuckle joint at a fully closed value, and because no component in the recording path issued a gripper command, the hand remained closed for the entire duration of every clip while the grasp was represented only by the attached marker.

This was not caught by the object-attachment check, which passed: the marker genuinely was following the gripper. It was following a gripper that never opened. (*Simulated.*) The recording path now commands the gripper controller directly, closing to a width derived from each object’s dimensions rather than to full closure, and the marker attaches only while the measured joint indicates closure on an object.

4.6 Clearance: a contact test is not a margin

Driving every task and measuring clearance to the wearer produced the result that has the largest consequence for the experimental programme. Every scenario in the chest-height working band passes inverse kinematics with collision checking enabled — the planner reports the poses valid — and yet the arms pass within 48–67 mm of the wearer’s torso, which is inside the 120 mm margin the real-robot configuration enforces. (*Simulated; per-run summary.json.*)

The reconciliation is that the two tests ask different questions. The planner’s check is binary contact; the floor is a margin. A scenario can satisfy one and be refused by the other, and this set of scenarios does.

A dedicated probe established three things about the cause. The closest link is the wrist rather than the forearm, which indicates that the targets are too close to the torso rather than that the path routes badly. Moving the working band forward does not help: at a forward offset that clears the floor, only one of six probed poses remains reachable, and beyond that nothing is reachable at all — clearance and reach are in direct conflict along that axis. Moving the band outboard does work: increasing the lateral half-span from 155 mm to 250 mm gives 134 mm of clearance with all six poses still reachable, and every value tested beyond that also clears. (*Simulated; scripts/probe_task_band_clearance.py.*)

The corroboration is independent: the two tasks that already work at a lateral half-span of 400 mm measured 158–187 mm of clearance across thirty runs, with none below the floor.

The change was recorded and not applied, because a 250 mm half-span implies a 500 mm object, which exceeds the retention threshold of the compliant carrier used in one task and rescales the tilt threshold of another.

4.7 Verified task scenarios

All seventeen scenario definitions, and all seventeen coordinates written in the task protocols, were verified at ten repeats over paths densified to 20 mm, and all pass. (*Simulated; recordings/baselines/scenario_audit.yaml*)

Reaching that state required correcting figures that had been established by weaker checks. Eleven of the fourteen coordinates originally written into the protocol documents were reachable by no arm at all, being from a superseded table-height layout. One task’s documented hold and release poses were wrong in every particular, including the assignment of arms.

The specific failure mode is worth stating because it recurs: a pose at the edge of the feasible set passes intermittently. One pose measured 72–82% feasible per call, which passes a five-repeat check roughly one time in five.

4.8 Unit tests

Twenty-eight test files are present. Those encoding measured properties include the inverse-kinematics guard convergence, known-answer recovery for every continuous task metric, the discrete task outcome, and the cross-

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only

arm interference fit. Style tests (flake8, pep257) fail, and failed prior to this work, because the codebase uses

CONFIRM: double quotes throughout; this is pre-existing and is not a regression.

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Chapter 5

Discussion

5.1 What the geometry decides

The most consequential finding of this project is not a control result but a geometric one: the two arms, as mounted and as parked, share no reachable workspace. That single fact removes an entire class of task from the programme, and it does so in a way no amount of planning, control or autonomy can address, because the constraint is on where the arms can be at all rather than on how they get there.

It is worth being precise about what follows from it and what does not. Tasks requiring the two arms to meet at a point are blocked. Tasks requiring two distinct points a fixed distance apart are not, and were shown to be feasible in a band at chest height. The distinction was initially missed, and a bimanual programme was briefly declared blocked on the strength of a test that answered the stronger question. The correction matters more than the original error: the requirement a task actually imposes should be tested, rather than a convenient proxy for it.

5.2 Clearance and reach are in conflict, and the conflict is structural

The clearance result in Chapter 4 deserves emphasis because it is easy to read as a tuning problem. It is not. The planner's collision check and the deployed clearance floor ask different questions — contact versus margin — and a scenario can satisfy the first while being refused by the second. Every chest-band scenario in this project is in exactly that position.

Attempting to resolve it by moving the working volume forward fails, and the failure is instructive: clearance improves and reachability collapses, because the volume moves towards the edge of the arm's reach as it moves away from the wearer's body. The axis along which the conflict can be resolved is lateral, not fore-and-aft. That was not obvious in advance and was found only because the probe measured which link was closest, rather than only how close it was.

The ethical position is one-directional. The floor exists to protect a person who did not choose the motion, so the task moves and the floor does not. That constraint is recorded rather than negotiated.

5.3 Instrumentation failed more often than the system

A recurring pattern across this project, and one which shaped its methodology more than any design decision, is that surprising measurements were more often faults in the measuring instrument than in the system being measured.

The instances are numerous enough to be a finding rather than an anecdote. A noise floor of zero on every sensor channel was the recorder oversampling a slower sensor. A detection rate near zero was a renderer producing images outside the detector’s training distribution. A workspace statistic that varied by a factor of two between identical runs was a summary statistic sensitive to a rare tail event. A set of verification clips reported as failing were failing a colour test calibrated against the requested marker colour rather than the rendered one. A carried object reported as stationary was a centroid dragged towards fixed overlay text. In each case the system was behaving correctly and the instrument was not.

The methodological response — requiring every analysis routine to recover a known answer before it is trusted with real data — is, in the author’s view, the most transferable contribution of the work. It is also incomplete: the rule was adopted partway through and the earlier measurements have not all been re-derived under it.

5.4 Silent failure as the dominant failure mode

The safety work in this project consistently found that the problem was not that mechanisms failed to act, but that acting and not acting looked identical from outside. The dead-man defect is the clearest example: the mechanism was correct, its trigger condition was correct, and it could not fire on the failure that actually occurs, because the upstream node degraded by republishing stale data rather than by falling silent. Every test of it had passed, because every test removed the cable.

The general lesson is that a test which exercises a failure mode different from the one that occurs provides confidence without providing safety. The system now distinguishes data that is absent from data that is stale, and treats a guard that has stopped reporting as being in an unknown state rather than a clear one.

5.5 Why no end-to-end learned policy

A vision–language–action model was considered and deliberately not used, for three reasons that are worth recording because the decision would otherwise look like conservatism.

First, embodiment: available models are trained on datasets that contain no backpack-mounted supernumerary configuration, and there is no demonstration data for this platform with which to fine-tune. Second, and decisively, such a model emits actions directly, whereas every safety mechanism in this system sits between a commanded *pose* and the arm. A policy emitting joint targets bypasses collision-aware inverse kinematics, the clearance floor and the step guard. On a robot mounted beside a person’s head that is the wrong trade. Third, a failure in the modular pipeline is attributable to a named stage with its inputs logged, whereas a failure of a learned policy is a number that was wrong.

5.6 Limitations

5.6.1 Nothing has been validated on hardware

The most serious limitation is stated plainly: no result in this thesis was obtained from a physical arm, a physical master or a human participant. The simulation uses a mock hardware interface that echoes commands with no dynamics, which means it cannot answer questions about payload, compliance, latency under load, or any effect of real actuator behaviour. Where the simulation has been asked such a question it has returned a null result that is a property of the mock rather than evidence about the robot.

5.6.2 The input is degraded and the baseline depends on it

Seven of fourteen master-arm channels are unreliable. Every direct teleoperation condition in the experimental design depends on those channels, so the baseline against which autonomy would be compared is currently compromised. Repairing two specific channels was identified as the highest-value intervention.

5.6.3 The experimental programme is designed but unexecuted

The protocols, measures, counterbalancing and analysis routines exist and have been exercised with scripted input, but no participant has taken part. The hypotheses are pre-registered intent.

5.6.4 The commit history is not a development record

The project's version control history consists of one large initial import and a single day of subsequent commits, so it cannot support claims about the order or timing of development. The dated engineering log is the usable chronology, and it is a laboratory notebook rather than an independent record.

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Chapter 6

Conclusion

This project produced a complete, safety-instrumented teleoperation stack for a backpack-mounted pair of seven-degree-of-freedom arms commanded by a remote operator, together with a characterisation of what that configuration can do.

The engineering contributions are the pipeline itself — master-arm sensing with a spherical pose decomposition, collision-aware inverse kinematics with redundancy resolution, a clearance margin referenced to the wearer’s body, and a multi-path emergency stop whose blocking behaviour is made externally visible — together with a degraded-operation mode that keeps the system usable on a subset of its intended sensing and refuses to invent a position when none survives.

The measured contributions are a workspace characterisation of the mounted configuration, including the negative result that the two arms share no reachable workspace and that inter-arm handover is therefore blocked by geometry; a quantified account of what degraded sensing costs; the finding that the deployed clearance margin and the planner’s collision test disagree across the entire working band, and that the conflict is resolvable laterally but not fore-and-aft; and a set of eighty-seven recorded verification runs with automated checking of object handling, gripper actuation and wearer clearance.

The methodological contribution is the discipline of validating an instrument against a known answer before reporting a surprising measurement taken with it. This arose from necessity rather than principle: across the project, unexpected results were more often defects in measurement than in the system, and the practice of requiring every analysis routine to recover a constructed answer before being trusted with real data caught a substantial number of errors that would otherwise have entered the record.

6.1 Future work

The immediate next steps are determined by the limitations rather than by ambition. Two specific master-arm channels should be repaired, since the analysis identifies them as the constraint on the left arm’s usable motion and every direct-teleoperation baseline depends on them. The clearance conflict should be resolved by relocating the working band laterally, with the object specifications re-derived and the scenarios re-verified in one deliberate pass. The real-arm bring-up should be repeated with logging retained, so that the control rate, homing behaviour and session handling become citable rather than recorded only in a notebook.

Beyond that, the experimental programme is designed and instrumented but unexecuted, and executing it is the substantial remaining work.

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Chapter A

Repository Structure

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Chapter B

Verified Task Scenarios

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Chapter C

Verification Recordings

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Bibliography